DESCENT FOR MODULES ALONG A HYPERCOVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stub lemma-hypercovering-descent-sheaves (registered card L46299); module pullback a^* = a^{-1}(-) tensor over a^{-1}O_C from the chapter's notation.

Editorial note. The label lemma-hypercovering-descent-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Descent for sheaves along a hypercovering. Same unit, Lemma lemma-hypercovering-descent-sheaves. a^{-1} is fully faithful with essential image the cartesian sheaves, and a_* is the quasi-inverse - the sheaf-level conclusion being transferred.
- Module pullback definition. Same unit, chapter notation for a^* on modules. a^*F = a^{-1}F tensor over a^{-1}O_C with O; the tensor factor collapses when a^{-1}O_C = O.

Source reference. Original reference 0D8E: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 4167–4167. Prerequisite pointer invoked at proof line(s) 4706; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a site with equalizers and fibre products. Let $\mathcal{O}_\mathcal{C}$ be a sheaf of rings. Let K be a hypercovering. With notation as above*

$$a^* : Mod(\mathcal{O}_\mathcal{C}) \to Mod(\mathcal{O})$$

is fully faithful with essential image the cartesian $\mathcal{O}$-modules. The functor a_* provides the quasi-inverse.

Proof. Since $a^{-1}\mathcal{O}_\mathcal{C} = \mathcal{O}$ we have $a^* = a^{-1}$. Hence the lemma follows immediately from Lemma 0D8E. □

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